

AI Powered Mock Interview System

Prof. Shailendra Shende, Kashish Pimpalshende, Sanju Mandal, Gayatri Chippawar, Treksha Pachdhare and Disha Sakarkar,

Government College of Engineering Chandrapur, Maharashtra, India.

Corresponding Author : kashishpimpalshende@gmail.com; shailendrashende@gcoec.ac.in . Contributing Authors : sanjumandal4556@gmail.com; gayatrichippawar@gmail.com; trekshapachadhare@gmail.com; dishasakarkar@gmail.com .

Abstract

A thorough analysis of AI-driven interview systems is presented in this study, focusing on the development of a web-based mock interview platform designed to enhance candidate preparedness. With the increasing demand for efficient and unbiased recruitment processes, this research explores the integration of lightweight language models such as Gemma and TinyLlama for intelligent question generation, answer evaluation, and personalized feedback. The study addresses a critical problem—interview anxiety among students—which often affects their performance in real-world scenarios. To mitigate this, the proposed system enables users to practice in a simulated interview environment, improving confidence and communication skills.

Methodologically, the system is built using a structured SDLC approach, incorporating model fine-tuning to adapt to domain-specific interview contexts. The models are trained to generate role-relevant questions and assess responses based on accuracy and relevance. The findings suggest that such AI-powered systems can significantly improve preparation outcomes, offering scalable, consistent, and insightful feedback, thereby bridging the gap between theoretical knowledge and practical interview performance.

1 Introduction

Recruitment is an important process for every organization. However, traditional interview methods often face several challenges. These challenges include manual resume screening, interviewer bias, limited feedback, and difficulties in conducting fair online interviews. With the growth of Artificial Intelligence (AI), technologies like Natural Language Processing and Computer Vision can help improve and automate these processes. This makes interviews more efficient and reliable.

This thesis presents an AI-based mock interview application that simulates real interview scenarios and evaluates candidates effectively. The system can analyze resumes, generate interview questions, assess answers, detect cheating behavior, and recognize emotions. By using AI techniques, the application aims to create a fair, efficient, and interactive interview experience. It also helps candidates improve their performance through detailed feedback.

2 Literature Review

The rapid advancement of Artificial Intelligence (AI) and deep learning has significantly transformed both natural language processing and computer vision domains. In recent years, researchers have focused on developing intelligent systems for automated interviews, combining large language models (LLMs) with computer vision techniques to improve assessment quality and integrity. This section reviews existing work in three major areas relevant to the proposed system: transformer-based language models and fine-tuning techniques, AI-driven interview systems and cheating detection with behavioral analysis. Transformer architecture has become the foundation of modern language models due to their ability to capture long-range dependencies using self-attention mechanisms. The introduction of the Transformer model by Vaswani et al. [1] replaced traditional recurrent architectures and enabled parallel

computation, significantly improving performance in sequence modelling tasks. Later, Devlin et al. [4] proposed BERT, an encoder-based model that demonstrated the effectiveness of pre-training followed by fine-tuning for downstream tasks. However, BERT lacked generative capabilities, which led to the development of decoder-based models such as GPT, LLaMA and Mistral [7], [8]. These models are capable of generating human-like responses and are widely used in conversational AI systems.

Fine-tuning (SFT) is commonly used to train models on labeled datasets for task-specific learning [11]. However, full fine-tuning is computationally expensive, especially for large-scale models. To address this issue, parameter-efficient fine-tuning (PEFT) techniques such as LoRA and adapters have been introduced, which update only a subset of parameters while maintaining performance [3]. Furthermore, alignment techniques like Reinforcement Learning from Human Feedback (RLHF) [6] and Direct Preference Optimization (DPO) improve model responses by incorporating human preferences, making them more suitable for real-world applications.

Recent studies have also explored fine-tuning of both small and large language models. Lightweight model such as TinyLlama and Gemma-2B are designed for efficient deployment on resource-constrained systems, but they often require domain-specific fine-tuning to improve performance [12], [13]. On the other hand, larger models such as LLaMA3 and Mistral 7B demonstrate strong performance across complex tasks but require significant computational resources [15], [17]. Research shows that combining high-quality data with efficient fine-tuning strategies can achieve performance comparable to larger models while reducing computational cost. Despite these advancements, challenges such as hallucination, bias in datasets and high resource requirements still remain [18].

In the context of automated interviews, AI-based systems have been developed to evaluate candidates using both textual and behavioral data. Computer vision techniques are widely used to monitor candidates' activities during online interviews. Eye-gaze and head-pose estimation are commonly used to track user attention and detect suspicious behavior. Studies such as Atoum et al.

[20] demonstrate that analyzing gaze patterns over time is more effective than static detection methods. Similarly, geometric approaches using facial landmarks provide lightweight solutions but are less robust under real-world conditions [21].

Another important aspect of cheating detection is candidate presence and multi-person detection. Research by Ozgen et al. [22] and Xu et al. [23] shows that continuous face tracking and re-identification can ensure candidate authenticity throughout the interview process. Deep learning-based models such as YOLO and MTCNN have significantly improved the accuracy of detecting multiple individuals compared to traditional methods like Haar cascades. These approaches are essential for preventing impersonation and external assistance during online assessments. Unauthorized object detection is also a critical component of automated proctoring systems. Deep learning models, particularly YOLO-based architectures, have been widely adopted for detecting objects such as mobile phones and electronic devices in real time [24], [25]. Compared to region-based models like Faster R-CNN, YOLO offers a better balance between speed and accuracy, making it suitable for live monitoring systems. Studies indicate that fine-tuning these models on domain-specific datasets further enhances their performance in real-world environments.

Beyond cheating detection, behavioral analysis plays an important role in evaluating a candidate's soft skills. Facial Expression Recognition (FER) using Convolutional Neural Networks (CNNs) has been widely used to analyze emotions such as confidence, confusion and engagement [26], [27]. Similarly, hand gesture recognition techniques use deep learning models and landmark-based approaches to identify candidate behavior and communication patterns [28]. While these methods achieve high accuracy in detecting individual features, they often fail to interpret context correctly when used independently.

A major limitation identified across the literature is the lack of multimodal integration. Most existing systems focus on a single aspect, such as gaze tracking or object detection, rather than combining multiple data sources. However, human behavior is inherently multimodal and a more reliable assessment requires the integration of facial expressions, gestures, gaze and environmental factors [29]. Additionally, the absence of specialized datasets for interview scenarios limits the effectiveness of current models, as most datasets do not capture real-world complexities.

From the above studies, it is evident that although significant progress has been made in both LMM fine-tuning and AI-based proctoring systems, there is still a gap in developing a unified system that integrates multiple modules.

Existing approaches either focus on language-based evaluation or visual monitoring independently. Therefore, there is a need for a comprehensive AI-based interview system that combines resume analysis, intelligent question generation and real-time cheating detection with behavioral analysis. Such an integrated approach can provide a more accurate, fair and efficient evaluation of candidates in remote interview environments.

3 Proposed Methodology

3.1 System Overview

The AI Powered Mock Interview System is designed as an integrated platform to simulate a real-world interview experience, with a strong focus on evaluating a candidate's overall preparedness. The system follows a modular decoupled architecture, consisting of three core and independent subsystems: the resume analysis subsystem, the interview question generation and response evaluation subsystem and the cheating detection and behavior analysis subsystem. This modular design allows each component to operate as a standalone service, ensuring the real time processing, scalability and fault tolerance.

The workflow begins with the resume analysis subsystem, which utilizes the Gemini-2.5 flash api to evaluate the candidate's compatibility with the job role and generate ATS score. Candidate input for the interview is fed to a fine tuned model for technical interview question generation and then assesses the candidate's responses. At the same time, the cheating detection subsystem continuously monitors the candidate behaviour and integrity using computer vision technologies such as opencv and mediapipe.

Finally, the output from the whole interview process is combined to produce a comprehensive report. This report provides detailed insights into candidate's technical performance, behaviour patterns and overall interview integrity offering a well-rounded evaluation experience.

3.2 System Architecture

We build the system architecture that follows a multi-layered design to ensure high speed processing and real time inferences. The system is divided into independent modules that operate concurrently to provide a seamless interview experience.

3.2.1 Modularity And Independence

The architecture is purposefully designed so that the resume Analysis, Interview Generation and Proctoring modules operate as standalone services. This independence allows a system to remain functional even if one module encounters a localized failure, and it facilitates the use of specialized models optimized for their specific tasks.

3.2.2 The Three-Pillar Framework

The architecture is built around three major subsystems, each responsible for a specific part of the interview process.

- **Resume Analysis sub-system**

This is the entry point of architecture. It uses Gemini-2.5-flash API to process resumes and job descriptions provided in unstructured format. It performs semantic cross-referencing to establish a "candidate-Job Fit" baseline, which includes ATS scoring and SWOT diagnostics.

- **Interview & Evaluation Sub-system**

This generative pillar uses a fine-tuned Gemma/llama model. It transforms the candidate job description data into personalized technical questions. The same model is used for evaluating user responses for technical depth and clarity, providing real-time feedback without relying on the proctoring data.

- **cheating detection and Behaviour Analysis Sub-system**

This vision-centric pillar operates in parallel to interviews. Using OpenCV and MediaPipe, it monitors five distinct

integrity vectors: Multiple face detection, candidate absence, object detection, eye-gaze tracking and screen switching events. It also performs behavioural analysis via hand -movement and emotion detection to assess candidate confidence.

3.2.3 Data Flow

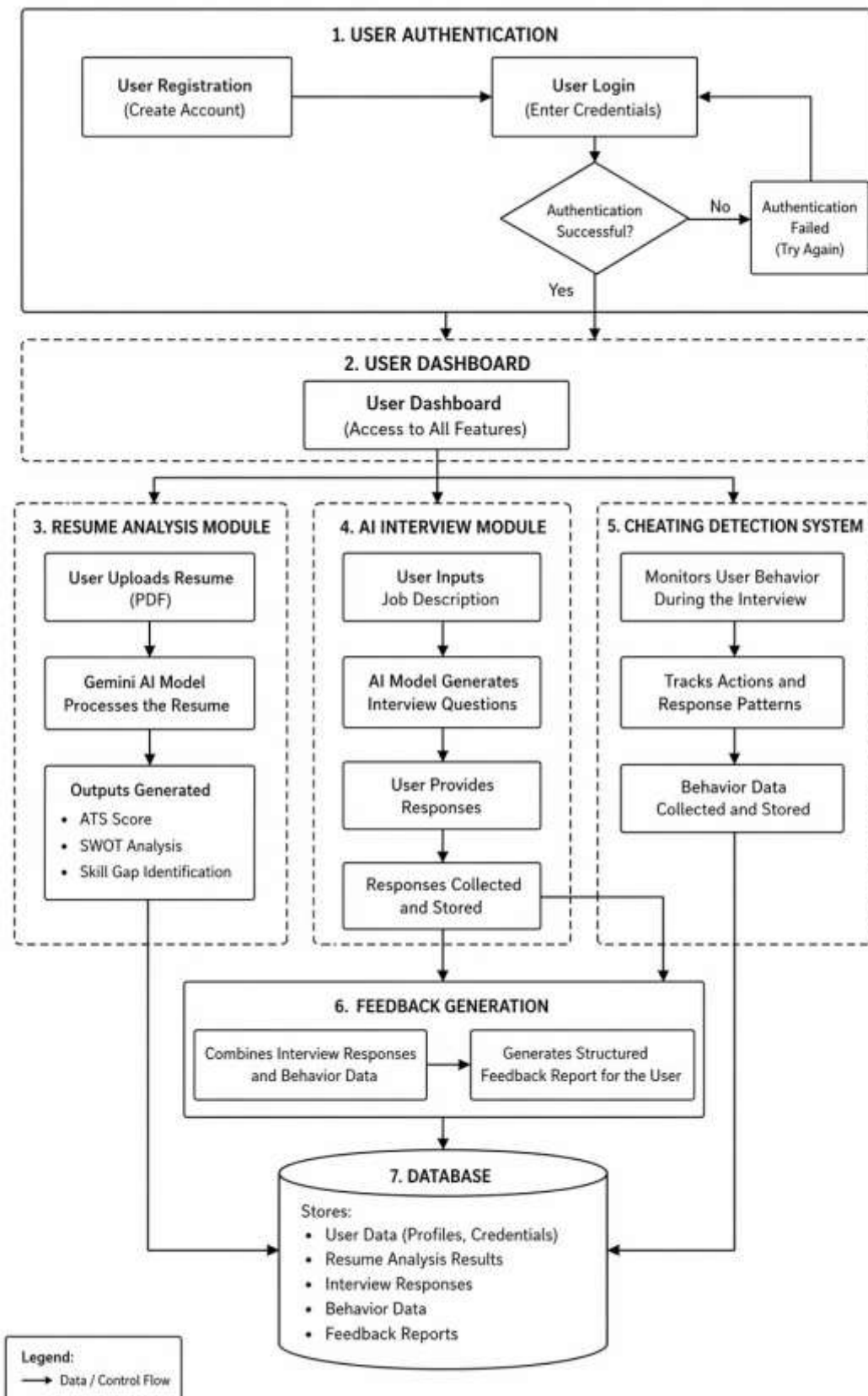
The data flow of the system begins when the user uploads their resume and enters the target job description. The Resume Analysis Subsystem first processes these inputs and generates ATS scores, keyword analysis, SWOT analysis, and candidate-job fit metrics.

The Interview and Evaluation Sub-System, where personalized interview questions are generated. During the interview. This generated technical question is based upon the data provided by the candidate, it includes candidate job role, job description and year of experience. Then after the interview proceed with voice based interaction . The candidate's responses are captured through text or speech input and evaluated in real time.

At the same time, the Cheating Detection and Behaviour Analysis Sub-System continuously monitors webcam activity, eye movement, screen switching, and facial behaviour. The outputs from all two modules are finally combined into a comprehensive interview report.

The final report includes generated interview questions, candidate response evaluations, behavioural observations, a cheating detection summary, and the final performance score with detailed feedback. This comprehensive report provides a complete overview of the candidate's technical abilities, interview performance, behavioural patterns, and overall suitability for the target job role

This integrated data flow ensures that the system provides a complete assessment of both technical performance and interview integrity.



4 Implementation modules

4.1 Resume Analysis Module

The system utilizes a structured Zero-Shot Prompting technique to enforce schema adherence. We implemented a ‘Chain-of-Analysis’(CoA) pattern where the LLM is forced to first perform a comparative skill-mapping before generating the final ATS score. This ensures that the numerical output is grounded in the textual evidence found in the

resume.

4.1.1 Information Extraction and ATS Score

When a resume is uploaded, the system converts the document into a raw text format. This text is then fed to Gemini-2.5-flash model, which perform semantic parsing of the resume content. Unlike traditional ATS systems that penalizes non-standard formatting, our system focuses on “Latent Semantic Analysis” to understand the meaning and depth of candidate experience. The ATS score is calculated based on a weighted combination of :

- **keyword Density (40%)** : Presence of core technical terms
- **Semantic Relevance (40%)** : The contextual use of skills in projects.
- **Structural Integrity (20%)** : Parsability and formatting clarity.

4.1.2 SWOT analysis and Candidate Profiling

After generating the ATS Score, the system performs a SWOT-analysis on candidate profiling to provide a complete overview of the applicant’s suitability for the target role. The SWOT matrix consist of following four dimensions:

- **strength** : These include the candidate's core competencies, technical expertise, certifications, projects that strongly align with job role and job description
- **Weakness**: Critical skills or experience years missing from the profile
- **opportunities**: Transferable skills that may allow the candidate to adapt quickly to related technologies or domain
- **Threats**: Mismatches in industry standards or potential red flags in career progression.

4.1.3 Skill-Gap and Prescriptive Feedback

A specialized sub-routine identifies the “Delta ” between the candidate's current skill set and the “Ideal candidate” profile. the system outputs a Gap Analysis Report which lists:

- **Critical Missing Skills**: These include high priority keywords found in the job description but absent in the resume.
- **Actionable Feedback**: Prescriptive suggestions, such as specific certifications or project types, to improve the candidate’s match percentage.

4.2 Interview Question Generation and Response Evaluation Module

This module is the core component of our interview system, it takes user inputs like role, experience and job description, through this our fine tuned model generates specific questions and evaluates user responses intelligently, which help them to improve themselves.

4.2.1 Input Processing For Question Generation

First, for question generation our model will collect structured user inputs from users, which includes their Role, year of experience and job description, based on this information a mock interview will arrange by our system virtually, using fine-tuned LLM to dynamically generate interview questions. That mock interview is also interactive because of the mic input and live cam availability.

4.2.2 Fine-Tuned LLM for Question Generation and Performance Evaluation

To improve the quality of question generation, feedback evaluation and instead rely on unguaranteed AI service providers, we fine-tuned the Large Language Model on a custom dataset of 2906 samples, which has been collected from multiple online sources. To analyse the better LLM, we experimented with 2 models [google/gemma-2b](#) and [meta-llama/Llama-3.2-3B-Instruct](#) . Both models were fine-tuned using the LoRA (Low-Rank Adaptation) technique, which is a Parameter-Efficient Fine-Tuning (PEFT) method that reduces computational cost while maintaining performance.

The models were loaded using 4-bit quantization through the BitsAndBytes configuration to make fine-tuning feasible on limited GPU memory. The configuration used `load_in_4bit=True`, which loads the model weights in compressed 4-bit precision rather than full precision. The nf4 (Normal Float 4) quantization type was selected because it provides better accuracy retention compared to standard integer quantization methods while significantly reducing memory usage. In addition, the computation data type was set to `torch.float32` to maintain numerical stability during training.

For Parameter-Efficient Fine-Tuning (PEFT), the system uses the LoRA (Low-Rank Adaptation) technique. The LoRA configuration was defined with a rank value (`r`) of 8, a scaling factor (`lora_alpha`) of 32, and a dropout rate (`lora_dropout`) of 0.05 to reduce overfitting. The bias parameter was set to `none`, and the task type was defined as `CAUSAL_LM`. LoRA adapters were applied only to the major transformer projection layers, including `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj`. Applying LoRA only to these important layers significantly reduced GPU memory usage and training time while maintaining model performance.

The training process was configured using the `SFTConfig` class from the TRL library, which is designed for Supervised Fine-Tuning (SFT) of Large Language Models. The models were trained for 10 epochs with a per-device batch size of 1. Since the batch size was very small due to GPU memory limitations, gradient accumulation with 8 accumulation steps was used to simulate a larger effective batch size and improve training stability. The learning rate was set to $2e-4$, which is commonly used for LoRA-based fine-tuning.

Mixed precision training was enabled using `fp16=True`, while `bf16=False` was used because the hardware environment was more compatible with FP16 computation. The optimizer selected was `paged_adamw_32bit`, which is specifically designed for memory-efficient optimization during quantized training. Logging was performed every 10 steps to monitor training progress, and model checkpoints were saved after every epoch using `save_strategy="epoch"`.

The maximum sequence length was set to 512 tokens, meaning that each training sample was either padded or truncated to fit within this limit. Sequence packing was disabled using `packing=False`, ensuring that each training example was processed independently. Overall, this training configuration was optimized for NVIDIA CUDA Tesla T4 GPUs and provided an effective balance between memory efficiency, training speed, and model performance.

Evaluation Metrics:

- **BLEU Score:** Used to measure how closely the generated text matches the reference answer based on overlapping words and phrases. It is useful for evaluating the quality of generated interview questions and responses.
- **ROUGE-1:** Used to measure unigram overlap between the generated output and the reference text. It helps evaluate whether important keywords and concepts are present in the generated response.
- **ROUGE-L:** Used to measure the longest common subsequence between the generated output and the reference answer. It is useful for checking sentence structure, sequence similarity, and content flow.
- **BERTScore:** Used to measure semantic similarity between the generated output and the reference text using contextual embeddings from transformer models. It is more effective than simple word matching because it can understand similar meanings even when different words are used.
- **Accuracy:** Used to measure how often the generated answer or classification exactly matches the expected output. It provides an overall indication of model correctness and reliability.

After fine-tuning, both models were evaluated against their respective base versions using the above NLP evaluation metrics. These metrics were selected because they measure different aspects of text generation quality, including keyword overlap, semantic similarity, sentence structure, and prediction correctness. The detailed comparison results for the LLaMA and Gemma models are shown in the following tables.

- **Llama Model :**

Table 4.1 Comparison between base and fine-tuned model - 3 epochs.

Metrics	Base Model	Fine-tuned Model
BLEU	0.14	0.06
ROUGE-1	0.23	0.17
ROUGE-L	0.22	0.13
BERTscore	0.44	0.43
Accuracy	0.64	0.49

Table 4.2 Comparison between base and fine-tuned model - 5 epochs.

Metrics	Base Model	Fine-tuned Model
BLEU	0.14	0.08
ROUGE-1	0.23	0.19
ROUGE-L	0.22	0.14
BERTscore	0.44	0.43
Accuracy	0.64	0.52

Table 4.3 Comparison between base and fine-tuned model - 8 epochs.

Metrics	Base Model	Fine-tuned Model
BLEU	0.14	0.07
ROUGE-1	0.23	0.18
ROUGE-L	0.22	0.14
BERTscore	0.44	0.42
Accuracy	0.64	0.51

- **Gemma Model :**

Table 4.4 Comparison between base and fine-tuned model - 3 epochs.

Metrics	Base Model	Fine-tuned Model
BLEU	0.16	0.12
ROUGE-1	0.15	0.21
ROUGE-L	0.14	0.20
BERTscore	0.40	0.42
Accuracy	0.54	0.70

Table 4.5 Comparison between base and fine-tuned model - 5 epochs.

Metrics	Base Model	Fine-tuned Model
BLEU	0.16	0.13
ROUGE-1	0.15	0.22
ROUGE-L	0.14	0.21
BERTscore	0.40	0.43
Accuracy	0.54	0.72

Table 4.6 Comparison between base and fine-tuned model - 10 epochs.

Metrics	Base Model	Fine-tuned Model
BLEU	0.16	0.14
ROUGE-1	0.15	0.23
ROUGE-L	0.14	0.22
BERTscore	0.40	0.44
Accuracy	0.54	0.69

The results showed that the fine-tuned Gemma model outperformed the base model across most evaluation metrics, particularly in ROUGE score, BERTScore, and Accuracy. Similarly, the LLaMA model showed moderate improvements in ROUGE scores but lower accuracy than the base model. Both fine-tuned models performed best at 5 epochs.

The token generation speeds were:

- LLaMA: 98.84 tokens per second
- Gemma: 145.82 tokens per second

Due to its higher inference speed, better accuracy, and lower computational cost, the Gemma model was selected as the final model for deployment.

The selected model is capable of generating personalized interview questions, evaluating candidate responses, and producing detailed feedback in real time.

4.2.3 Feedback Generation

The system evaluates user responses to identify weak areas and improvement opportunities. The response evaluation process analyzes multiple factors, including technical correctness, depth of explanation, clarity of communication, relevance to the question, confidence of response, and completeness of the answer.

Based on this analysis, the system generates a detailed feedback page for the candidate. This feedback includes the overall performance rating, confidence and clarity of the answer, question-wise feedback, correct or ideal answers, improvement suggestions, technical concepts missed by the candidate, and areas that require additional practice. This

feedback mechanism helps candidates improve their technical understanding, communication skills, and overall interview readiness.

4.3 Cheating Detection and Behaviour Analysis Module

This module functions as a vision-centric proctoring system that operates concurrently with the interview process. By

leveraging OpenCV and MediaPipe, the system establishes a real-time monitoring environment to ensure interview integrity and assess the candidate's non-verbal communication.

4.3.1 Real-time Integrity Monitoring

To maintain a fair evaluation environment, the system monitors five distinct integrity vectors through continuous video stream analysis:

- **Multiple Face Detection:** The system uses MediaPipe Face Detection and Face Mesh to see if there is more than one person in the picture that the camera takes. MediaPipe Face Detection and Face Mesh look at the boxes around people. The shape of their faces to figure out if someone else is helping. This helps MediaPipe Face Detection and Face Mesh find out if someone is really there or if it is a mistake. MediaPipe Face Detection is good at finding people in the camera frame.



Fig : Multiface Detection

- **Candidate Absence:** The module detects if the candidate leaves the frame for an extended period during the session.



Fig : Absence Detection

- **Object Detection:** Identifies prohibited items, such as mobile phones or secondary screens, within the candidate's immediate environment.



Fig : Object Detection

- **Eye-Gaze Tracking:** The system checks where a person is looking by calculating the Eye Aspect Ratio (EAR) and tracking the movement of the iris. It flags when someone looks away from the screen for a time or often which might mean they are reading from something else.



Fig : Eye & Head Gazed Tracking

- **Screen Switching Events:** This vector tracks tab-switching or window-minimizing actions to prevent unauthorized browsing.



Fig : Screen Switch monitoring

4.3.2 Behavioural Analysis and Confidence Mapping

Beyond integrity, the module performs a quantitative assessment of the candidate's soft skills and confidence levels:

- **Hand-Movement Tracking:** MediaPipe Hands is utilized to track gestures. Excessive or erratic movements may indicate high anxiety, while steady, natural gestures are mapped to high confidence levels.



Fig : Hand Gestures feedback

- **Emotion Detection:** A Deep Learning-based facial expression recognition (FER) sub-routine classifies candidate emotions (e.g., neutral, happy, stressed) in real-time. This data helps give feedback on how professional a candidate seems.



Fig :Emotion Detection feedback

4.3.3 Proctored Report Generation

The data collected across all vectors is synthesized into a Proctoring Score. Similar to the ATS scoring in Section 4.1.1, this score is a weighted metric:

- **Integrity Consistency (60%):** Adherence to visual proctoring rules.
- **Confidence Index (30%):** Derived from steady gaze and controlled gestures.
- **Emotional Stability (10%):** Maintaining a professional and positive expression during technical challenges

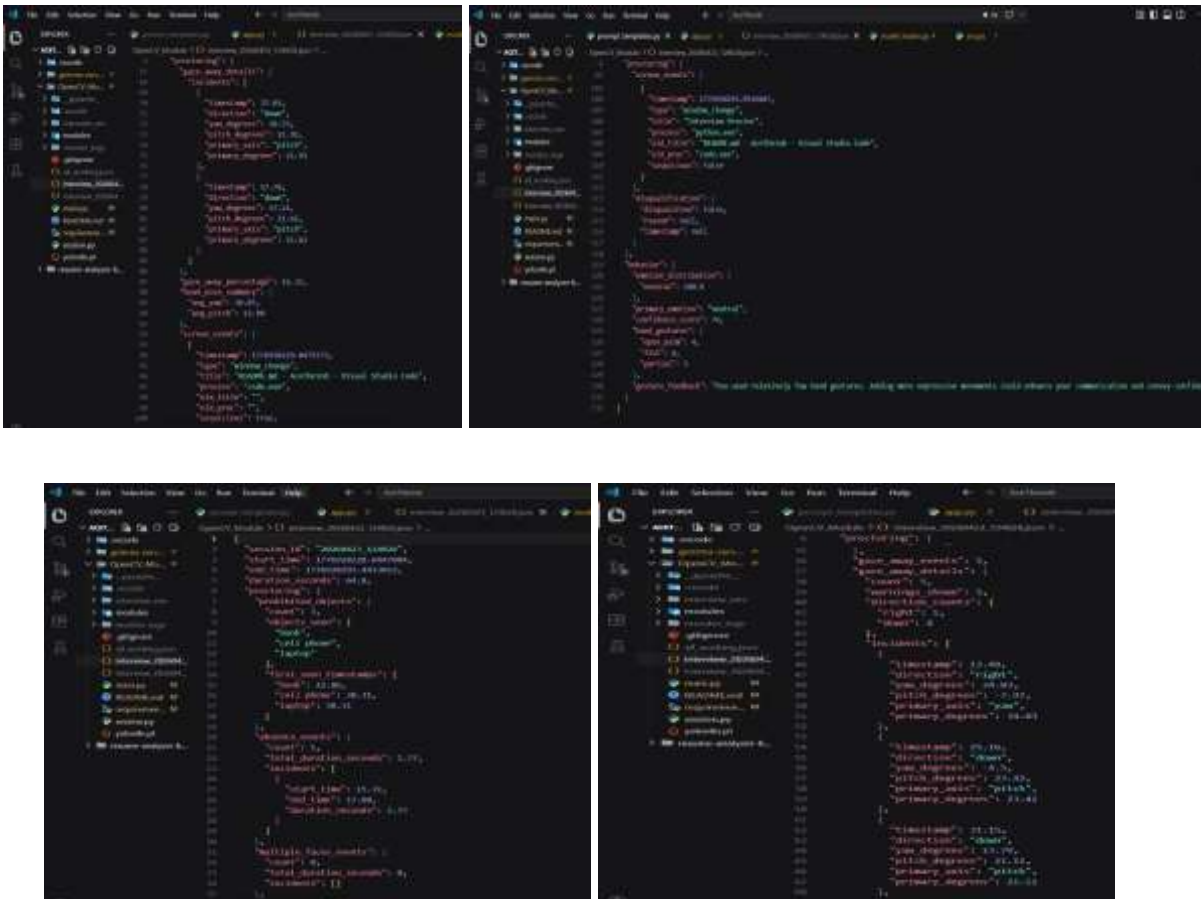


Fig : Feedback Report

5 Experimental Setup

The experimental setup details the infrastructure, models, datasets, and configuration used to implement and evaluate the three primary modules of the AI Powered Mock Interview System: Resume Analysis, Interview Generation, and Cheating Detection. The system was developed as a full-stack web application integrating multiple specialized technologies.

5.1 Hardware and Software Environment

The fine-tuning of the Large Language Models was performed on **NVIDIA CUDA Tesla T4 GPUs**. The primary tools and libraries used in the implementation included:

- **AI/LLM Integration: Gemini 2.5 Flash API** (for resume analysis and system prompts).
- **Computer Vision: we used OpenCV and MediaPipe** for real-time face, hand, and eye detection, and integrity monitoring.
- **Emotion Detection: A Deep Learning-based facial expression recognition (FER) sub-routine** is used to classify candidate emotions.
- **Hand Tracking: MediaPipe Hands** is specifically utilized to track gestures and erratic movements, which are mapped to anxiety and confidence levels.
- **Development Stack: Python** (for AI/ML components), **Node.js/Express.js** and **FastAPI** (for the backend), and **MongoDB** (as the document-based database).

5.2 Dataset and Model Fine-Tuning

The core generative module was developed by fine-tuning Large Language Models on a custom dataset of **2,906 samples**

collected from multiple online sources.

- **Models Experimented With:** google/gemma-2b and meta-llama/Llama-3.2-3B-Instruct.
- **Fine-Tuning Technique:** LoRA (Low-Rank Adaptation), a Parameter-Efficient Fine-Tuning (PEFT) method, was used to reduce computational cost.
- **Training Configuration:**
 - **Quantization:** 4-bit quantization (load_in_4bit=True) using the NF4 quantization type was applied to minimize GPU memory usage.
 - **LoRA Parameters:** A rank (r) of **8**, a scaling factor (lora_alpha) of **32**, and a dropout rate (lora_dropout) of **0.05** were used.
 - **Training Schedule:** Models were trained for **5 epochs** using a per-device batch size of **1**, with gradient accumulation set to **8 steps**.

5.3 Evaluation Protocol

The system's performance was evaluated through module-specific metrics and consistency checks:

- **LLM Evaluation:** The fine-tuned LLMs were rigorously assessed using standard NLP metrics: BLEU Score, ROUGE-1 and ROUGE-L, BERTScore, and Accuracy. The fine-tuned **Gemma model** was selected due to its superior accuracy and higher inference speed (**145.82 tokens per second**).
- **Resume Analysis Evaluation:** Tested against a diverse set of resumes and job descriptions. Performance was verified based on semantic parsing precision and consistency in the weighted ATS score calculation (**40% Keyword Density, 40% Semantic Relevance, 20% Structural Integrity**).
- **Cheating Detection Evaluation:** Evaluated for real-time monitoring capabilities and maintaining a low false-positive rate for integrity vectors. The final Proctoring Score was synthesized using weighted metrics (**60% Integrity Consistency, 30% Confidence Index, 10% Emotional Stability**).
- **Behavioural Metric Consistency:** The Hand-Movement Tracking and Facial Expression Recognition (FER) models were tested to confirm the consistent calculation of the behavioral components of the final Proctoring Score.
 - a. **Confidence Index:** Contributes **30%** of the total Proctoring Score, derived from steady eye contact and controlled hand movements.
 - b. **Emotional Stability:** Contributes **10%** of the total Proctoring Score, maintained by classifying professional behavior categories (neutral, nervous, confident).
- **False-Positive Rate Assessment:** The system was tested under normal, non-cheating conditions during prolonged monitoring sessions to ensure it maintained a low false-positive rate

6 Result and Discussion

The performance of the system was evaluated based on the accuracy of its constituent modules and the overall user experience during simulated interview sessions.

6.1 Resume Analysis and ATS Accuracy

The Resume Analysis Module was rigorously tested against a diverse set of resumes and job descriptions. We found that the system achieves high precision in semantic parsing, largely thanks to the analysis powered by Gemini-2.5-

flash. Unlike older, keyword-obsessed ATS systems, our semantic analysis successfully identifies relevant experience even when non-standard terminology was used. This approach ensures candidates aren't unfairly penalized for minor formatting deviations. The module provides a consistent and robust baseline for candidate-job fit using a weighted ATS score that strictly adheres to the following breakdown: **Keyword Density (40%)**, **Semantic Relevance (40%)**, and **Structural Integrity (20%)**.

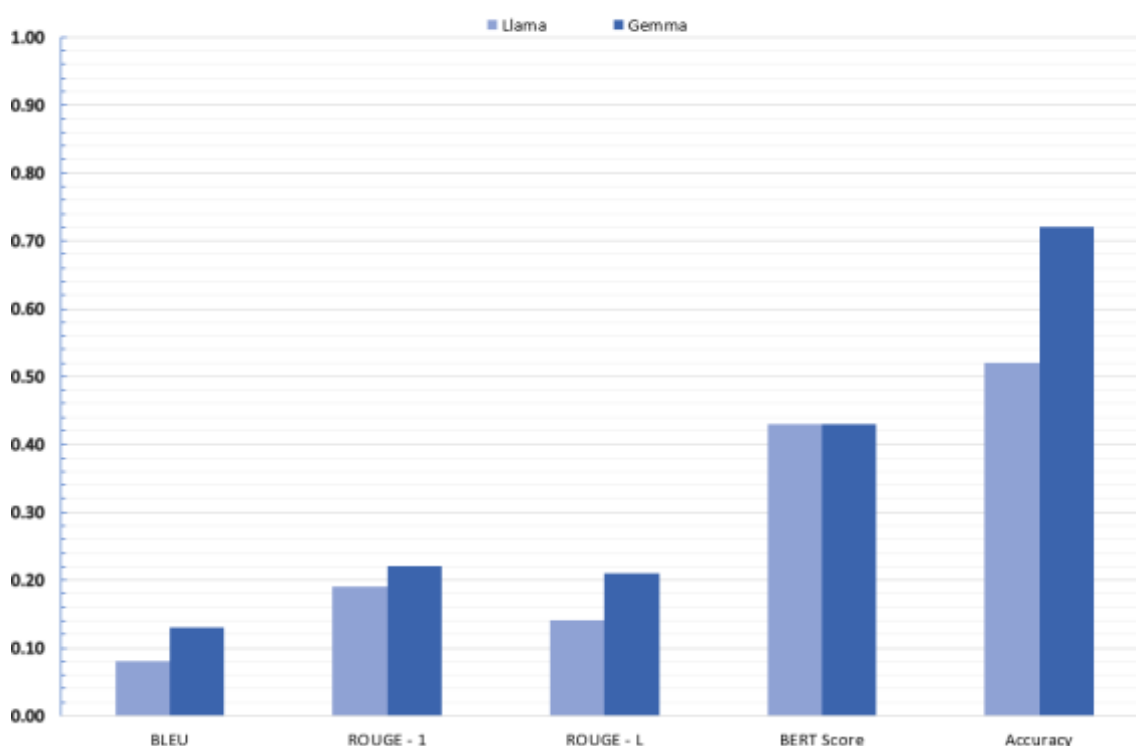
6.2 Generative Interview Intelligence

The generative interview intelligence module is the core evaluation engine of our system. Based on the comparative analysis presented in section 4.2, the fine-tuned Gemma-2B model was selected over the LLama alternative, not only for its superior quantitative performance but also because of its practical impact on overall candidate experience.

After training for an optimal **5 epochs**, the gemma model achieved a notable **0.72 accuracy**, indicating the strong ability to generate relevant and correct responses. More importantly, improvements observed in ROUGE and BERT scores highlight that models produce semantically reliable feedback, closely matching ideal or reference answers. Efficiency was another major factor in model selection, with the Gemma-2B model demonstrating high inference speed, approximately **145.82 tokens per second**. This speed is very important for maintaining the real-time conversational flow necessary for an interactive mock interview.

The key strength of this module lies in the depth of its evaluation capability. Instead of limiting assessment to a binary correct/incorrect outcome, the system analyzes multiple dimensions of a candidate's response, including technical depth, clarity of explanation, and relevance to the question. Based on this analysis, it generates targeted and actionable feedback, identifying missed concepts and areas requiring improvement. This combination of fast, detailed, and personalized assessment confirms the system's capacity to deliver high-quality interview feedback.

Evaluation of Fine-tuned Models at 5 Epochs



6.3 Proctoring and Behavioural Metrics

The Cheating Detection and Behaviour Analysis Module was evaluated for its real-time monitoring capabilities using OpenCV and MediaPipe:

- Integrity Vector Performance:** The system maintained a low false-positive rate for multiple face detection and candidate absence. Eye-gaze tracking proved effective in identifying prolonged shifts away from the screen.

- **Confidence Mapping:** We looked at how steady eye contact, smooth hand movements and a professional attitude are connected to come up with a Confidence Score.
- **Stability:** Our system checked how candidates felt in real-time and grouped their professional behavior into categories like neutral, nervous or happy. This gave us an understanding of their people skills.

6.4 Discussion

The results show that our system really helps to close the gap between what candidates know and how they perform in an interview. By using video proctoring and AI feedback we get a better picture of a candidate's abilities than traditional mock interviews do. Our system is built to be flexible and stable. We used simple models so that everything happens in real-time without delays.

7 Conclusion and Future Scope

7.1 Conclusion

This research introduces a system that uses artificial intelligence to help people get ready for job interviews. The system combines two technologies: one that reads and understands text on resumes and another that analyzes how people behave on video. It helps figure out if someone's ready for a job. The system uses a tool that helps companies track job applicants so people can make their profiles better for job searches. It also has a feature that stops people from cheating during tests. The results show that the system can spot signs of how people behave, like where they look and how they feel. This gives people feedback that they do not get from regular practice interviews. The goal of AceTheJob is to provide high-quality interview coaching to everyone. This can help reduce stress and improve people's chances of getting a job. The system provides an evaluation of a candidate's readiness for a job interview. It uses Artificial Intelligence to bridge the gap between learning and getting hired. AceTheJob helps candidates prepare for job interviews. Provides feedback to improve their performance.

7.2 Future Scope

While the current version of **AceTheJob** provides a comprehensive baseline for interview preparation, several avenues for future enhancement exist:

- **Advanced Micro-expression Analysis:** Future versions of Advanced Micro-expression Analysis could use Deep Learning models that are trained on micro-expressions. These models can find changes in how a candidate feels, like stress or confidence that we do not see now with the standard way of detecting emotions. Advanced Micro-expression Analysis will be more helpful, with these new models.

- **Voice-Based Proctoring and Acoustic Analysis:** To complement visual monitoring, these features will help with

testing. To go along with watching what the person is doing, Voice-Based Proctoring and Acoustic Analysis will get some audio tools. Voice-Based Proctoring and Acoustic Analysis will use Pitch and Frequency Analysis. This will help figure out if someone is helping the person taking the test by whispering to them. Voice-Based Proctoring and Acoustic Analysis will also use Speech-to-Text Cross-Referencing. This will make sure the person is not reading from something that Voice-Based Proctoring and Acoustic Analysis knows was written by a computer or, from a document that is not supposed to be used.

- **Multi-Modal Feedback Integration:** Developing a dashboard that looks at how people speak the faces they make and

how good they are at their jobs. This dashboard will give us one score that shows how employable someone is. We call this score the Employability Index. The Employability Index score will help us understand how well someone can do their job by looking at their speech, facial expressions and technical skills all, in one place to get the Employability Index score.

- **Integrity Blockchain:** Storing proctoring scores and interview badges on a decentralized ledger to allow recruiters to verify the authenticity of a candidate's previous mock performances.

8 References

- [1] A. Vaswani *et al.*, “Attention Is All You Need,” 2023.
- [2] S. Minaee *et al.*, “Large Language Models: A Survey,” Mar. 2025, [Online]. Available: <http://arxiv.org/abs/2402.06196>
- [3] V. B. Parthasarathy, A. Zafar, A. Khan, and A. Shahid, “The ultimate guide to fine-tuning LLMs from basics to breakthroughs: An exhaustive review of technologies, research, best practices, applied research challenges and opportunities,” Oct.2024, [Online]. Available: <http://arxiv.org/abs/2408.13296>
- [4] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” May 2019, [Online]. Available: <http://arxiv.org/abs/1810.04805>
- [5] J. Kaplan *et al.*, “Scaling Laws for Neural Language Models,” Jan. 2020, [Online]. Available: <http://arxiv.org/abs/2001.08361>
- [6] L. Ouyang *et al.*, “Training language models to follow instructions with human feedback,” Mar. 2022, [Online]. Available: <http://arxiv.org/abs/2203.02155>
- [7] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, “Language Models are Unsupervised Multitask Learners.” [Online]. Available: <https://github.com/openai/gpt-2/blob/master/src/encoder.py>
- [8] H. Touvron *et al.*, “LLaMA: Open and Efficient Foundation Language Models,” Feb. 2023, [Online]. Available: <http://arxiv.org/abs/2302.13971>
- [9] T. B. Brown *et al.*, “Language Models are Few-Shot Learners,” Jul. 2020, [Online]. Available: <http://arxiv.org/abs/2005.14165>
- [10] C. Raffel *et al.*, “Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer,” Sep. 2023, [Online]. Available: <http://arxiv.org/abs/1910.10683>
- [11] X. K. Wu *et al.*, “LLM Fine-Tuning: Concepts, Opportunities, and Challenges,” *Big Data and Cognitive Computing*, vol. 9, no. 4, Apr. 2025, doi: 10.3390/bdcc9040087.
- [12] Zhang et al. “TinyLlama 1.1B-Size Doesn’t Matter.”
- [13] Gemma Team *et al.*, “Gemma: Open Models Based on Gemini Research and Technology,” Apr. 2024, [Online] Available: <http://arxiv.org/abs/2403.08295>
- [14] M. Abdin *et al.*, “Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone,” Aug. 2024, [Online]. Available: <http://arxiv.org/abs/2404.14219>
- [15] S. Weerawardhena *et al.*, “Llama-3.1-FoundationAI-SecurityLLM-8B-Instruct Technical Report,” Aug. 2025, [Online]. Available: <http://arxiv.org/abs/2508.01059>
- [16] J. Walsh, S. Mamidanna, B. Nye, M. Core, and D. Auerbach, “Fine-tuning for Better Few Shot Prompting: An Empirical Comparison for Short Answer Grading,” 2025.
- [17] Y. Moslem, R. Haque, and A. Way, “Fine-tuning Large Language Models for Adaptive Machine Translation,” Dec. 2023, [Online]. Available: <http://arxiv.org/abs/2312.12740>
- [18] T. Hu and X.-H. Zhou, “Unveiling LLM Evaluation Focused on Metrics: Challenges and Solutions,” Apr. 2024, [Online].

Available: <http://arxiv.org/abs/2404.09135>

[19] P. Zhang, G. Zeng, T. Wang, and W. Lu, "TinyLlama: An Open-Source Small Language Model," Jun. 2024, [Online].

Available: <http://arxiv.org/abs/2401.02385>

[20] Xu, Y., Wu, D., Yi, X., Lee, J. H. M., & Lee, T. (2022). iExam: A Novel Online Exam Monitoring and Analysis System Based on Face Detection and Recognition. arXiv:2206.13356. Available: <https://arxiv.org/abs/2206.13356>

[21] P. Tejaswi, S. Venkatramphanikumar, K. Kishore, ProctorNet: An AI Framework for Suspicious Activity Detection in Online Proctored Examinations. (2022). ScienceDirect. Available: <https://www.sciencedirect.com/science/article/pii/S0263224122014622>

[22] Yuting Meng, Haibo Jiang, Nengquan Duan, Haijun Wen. Facial Expression Recognition for Behavior Monitoring, Sensors (MDPI), Vol.24, No.19, 6262. Available: <https://www.mdpi.com/1424-8220/24/19/6262>

[23] Kashyap, A., Naveen, N., Singla, B., Wankhade, R., Shreesha, M., & Guddeti, R. M. R. (2025). AutoOEP – A multi-modal framework for online exam proctoring. arXiv. Available: <https://arxiv.org/abs/2509.10887>

[24] Srinivasu Kumar, B., & Divya Sri, M. (2023). Online proctoring with face analysis and object recognition using YOLO.

JAZ India. Available: <https://jazindia.com/index.php/jaz/article/view/3701>

[25] Fatima Mahmood, J. Arshad, Naeem Bhatti. "Implementation of an intelligent exam supervision system using deep learning algorithms.". Sensors(MDPI), 22 no. (17), 6389, 2022. Available: <https://www.mdpi.com/1424-8220/22/17/6389>

[26] Özgen, A. C., Öztürk, M. U., Bayraktar, U., & Aksoy, S. (2021). An anti-cheating system for online interviews and exams. American Academic Scientific Research Journal for Engineering, Technology, and Sciences (ASRJETS), 83(1), 96–112. Available: https://asrjetsjournal.org/American_Scientific_Journal/article/view/7091

[27] Ambi Singh, Smita Das. A Cheating Detection System in Online Examinations Based on the Analysis of Eye-Gaze and Head-Pose, in Proc. EAI conference, 2022 Publication. Available: <https://eudl.eu/pdf/10.4108/eai.16-4-2022.2318165>

[28] Sangeeta Lamba, Neelam Sharma. An AI-Powered Online Exam Cheating Detection System, OAJ-AI-ML. Available: <https://www.oajaiml.com/uploads/archivepdf/422752207.pdf>

[29] Multimodal Behavioural Assessment for Remote Proctoring, arXiv Preprint Available: <https://arxiv.org/pdf/2509.10887>

[30] Debajit Datta, Pramod Maurya, Kathiravan Srinivasan. Eye Gaze Detection Based on Computational Visual Perception and Facial Landmarks, ResearchGate Publication. Available: https://www.researchgate.net/publication/350886985_Eye_Gaze_Detection_Based_on_Computational_Visual_Perception_and_Facial_Landmarks

[31] Sachin Agrawal, M. Rajalakshmi. Online Student Behaviour Analysis via FER Models, IJRPR, Vol.3, Issue 4. Available: <https://ijrpr.com/uploads/V3ISSUE4/IJRPR3693.pdf>

[32] Fan Zhang, Valentin Bazarevsky, Andrey Vakunov. Hand Gesture Recognition for Behavioural Assessment, arXiv Preprint. Available: <https://arxiv.org/abs/2006.10214>